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Technical Notes

Coupled-Mode Versus Single-Degree-of-Freedom Flutter: The Third Parameter in Flutter

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THE conventional flutter boundary uses two parameters, typically the dynamic pressure q and the Mach number M. This Note is inspired by the observation in computations and wind tunnel experiments that there can be a precipitous drop in the flutter dynamic pressure as the Mach number moves into the high subsonic/transonic range. For example, see Fig. 1, which is kindly provided by Dr. Walter Silva of the Aeroelasticity Branch at NASA Langley Research Center. But there is a third parameter: flow density, or, in nondimensional terms, a fluid/structure mass ratio. In the present brief Note, we explore the consequences of this third parameter for interpreting the results of flutter analysis, wind tunnel testing, and flight testing. This is of special importance when there is a transition from coupled-mode flutter to single-degree-of-freedom flutter (or conversely) as the Mach number moves through the transonic range. In particular, an explanation for the relatively poorer agreement between theory and experiment in the transonic regime will be offered. Also, the consequences of the third parameter will be explored for their significance in better understanding the distinction between coupled-mode flutter (also sometimes called bending torsion flutter or coalescing frequency flutter) due to negative aerodynamic stiffness (or its inverse, i.e., compliance) and single-degree-of-freedom flutter due to negative aerodynamic damping.

I. Two-Degree-of-Freedom Coupled-Mode Flutter Versus Single-Degree-of-Freedom Flutter

For two-degree-of-freedom flutter that is due to a loss of stiffness, leading to merging frequency flutter, the conventional (*two-parameter*) flutter boundary of q (dynamic pressure) versus M is useful. This is because the effect of the ratio of fluid to structural mass is small. However, for single-degree-of-freedom flutter due to a loss of damping in the system as a consequence of negative aerodynamic damping, the plot of q versus M is at best incomplete and often misleading.

By contrast, for single-degree-of-freedom flutter the most appropriate *two-parameter* surface is a reduced flow velocity, i.e., flow velocity divided by wing chord times a characteristic frequency (e.g., pitch natural frequency) versus Mach number.

However, the greatest insight is obtained by recognizing that the flutter boundary for both single-degree-of-freedom flutter and coupled two-degree-of-freedom flutter is a three-dimensional parameter surface. This has been explored and exploited in an earlier paper by Kholodar et al. [1]. When there is a transition from coupled-mode flutter to single-degree-of-freedom flutter or vice versa, it is especially important to keep all three parameters in mind.

The work in [1] was inspired by wind tunnel experiments done by the NASA Langley Research Center Aeroelasticity Branch [2], which have continued to be a source of further studies to this date.

II. Implications of a Three-Parameter Flutter Surface

There are several implications of this three-parameter flutter boundary surface, and two will be highlighted next.

First, if single-degree-of-freedom flutter occurs in a wind tunnel test, the dynamic pressure for flutter will drop rapidly in the transonic range, and it is better to test at constant q and varying M. Indeed, the (inviscid, zero structural damping) flutter q may drop to zero. See, e.g., the recent computational study by Stanford and Jacobsen [3] as well as the results in [1]. Adding structural damping or flow viscosity will increase this zero flutter dynamic pressure value, but given the uncertainty in either structural damping or the modeling of viscous effects in a Reynolds-averaged Navier–Stokes (RANS) code or any computational fluid dynamics code for that matter, there will always be greater sensitivity of the flutter boundary for single-degree-of-freedom flutter than for coupled-mode flutter. This has important implications when comparing theory and experiment and also in choosing flutter margins, given the greater uncertainty in identifying the flutter boundary for single-degree-of-freedom flutter.

The second point is that the test trajectory in three-parameter space is different in a wind tunnel than in flight. This is discussed in more detail in [1]. Perhaps the key point is that, for both the flight trajectory and the wind tunnel trajectory, the flutter boundary for single-degree-of-freedom flutter is more a Mach number boundary than a dynamic pressure boundary. Of course, the dynamic pressure is important in the following sense: The system may be unstable at very small (even zero) dynamic pressure, but the growth rate of the instability will be a function of dynamic pressure; i.e., the growth rate will be very small for very small dynamic pressure. This means that it will be more difficult to identify the single-mode flutter boundary than a coupled-mode flutter boundary in either a physical experiment or a time-marching computer simulation.

III. Choosing the Three Governing Parameters

There is no unique choice of the three parameters, and indeed the best choice varies depending on whether the phenomenon of interest is coupled-mode flutter or single-degree-of-freedom flutter. In dimensional terms, for a fixed set of structural parameters, the three flow parameters might be dynamic pressure, Mach number, and flow density. But equally one might use flow velocity (or, for that matter, speed of sound) rather than dynamic pressure.

Of course, these three parameters are related to each other in either a wind tunnel test or a flight test, but the relationship among them is different in a wind tunnel test than in a flight test. That is one reason why a wind tunnel test result is not the same as what will be obtained in a flight test. In a computational model, all three parameters are independent, which is why a computational model may provide results that can be compared to either a wind tunnel test or a flight test, provided that the appropriate "match" or relationship among the three parameters is used for the flight test or wind tunnel test.

More explicitly, consider first a wind tunnel test. Such tests are typically done by varying the Mach number and stagnation pressure, with the stagnation temperature held relatively constant. At a given Mach number, the static speed of sound is constant if the stagnation temperature is constant. But the dynamic pressure varies with the

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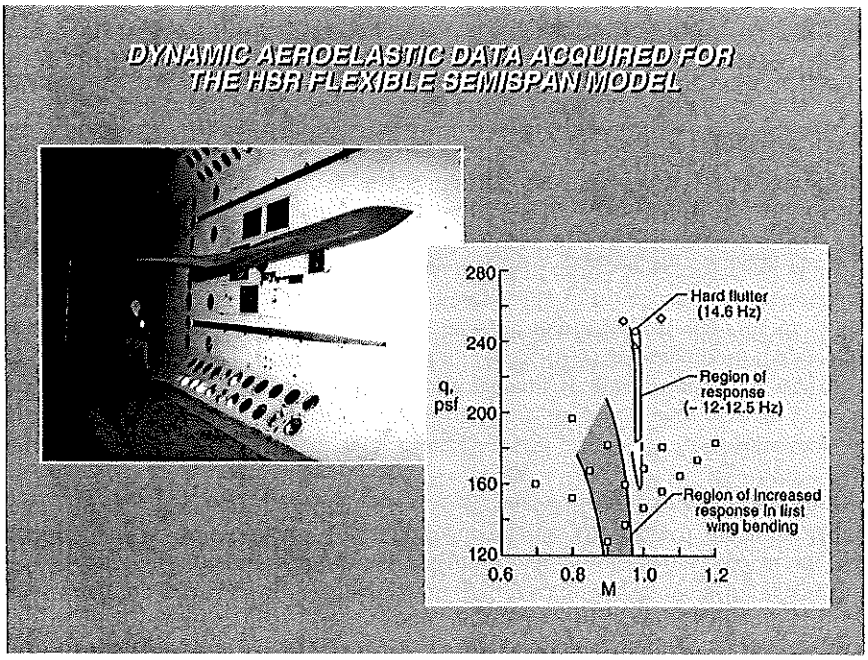


Fig. 1 Wing response over a range of dynamic pressure q and Mach number M.

Mach number even if the stagnation pressure is held constant and varies more if the stagnation pressure is changing. And, of course, the flow density is changing as well as the flow velocity.

On the other hand, in a flight test at a given altitude (and hence essentially constant temperature, speed of sound, and density), with a varying Mach number, the flow velocity is proportional to the Mach number, and the dynamic pressure varies as the square of the Mach number.

But in a computation, the analyst is free to choose the trajectory in the three-parameter space. So which one to choose? To address this question, it is useful and insightful to consider *three nondimensional parameters*, and this will be done next.

IV. Three Nondimensional Parameters

One of the parameters will be the Mach number, of course. The other two parameters might be the ratio of fluid to structural mass and the ratio of flow dynamic pressure to structural stiffness. Again, this is not the only choice of the three parameters, but it is a good choice; moreover, other useful choices may be derived from these three choices, or really from two of the parameters, since the Mach number is almost always the best choice for one of the three parameters.

[As a brief aside, we note that in low-speed flow, for $M \ll 1$, the Mach number may be ignored. Also, in some cases, the Reynolds number Re may be important as a further nondimensional parameter. For fluid structure interaction studies, it is the ratio of the boundary-layer thickness to the wing chord or plate length that is important. Of course, this ratio depends on Re. More will be said of this later.]

The specific choice of the mass ratio parameter will be the following:

$$\mu \equiv \rho_\infty L/m \tag{1}$$

where ρ is the flow density, L is the characteristic length (wing chord or plate length), and m is the mass per unit area of the structure. For a structure of variable mass, this might be the average mass per unit area. This definition of mass ratio μ is the inverse of that historically used in aeroelasticity. But, as we will see, the limit of zero fluid/structural mass ratio is meaningful, and it is more convenient to consider that limit than the limit of infinite structural/fluid mass ratio.

For the stiffness or compliance ratio, the following choice is made:

$$\lambda^* \equiv \left(\frac{U_\infty}{\omega_R L}\right)^2 \mu \tag{2}$$

where U_∞ is the flow velocity, L is the characteristic length, ω_R is the reference frequency (a resonance frequency of the structure), and, of course, μ is the mass ratio previously defined. This definition of compliance ratio has several interesting and useful properties. The product of flow density times flow velocity squared can be interpreted as flow dynamic pressure (with a factor of 2 inserted). Also, for coupled-mode flutter, this compliance ratio is of order 1, i.e., more than 0.1 and less than 10, as a scaling analysis of the usual (linear) aeroelastic equations of motion would suggest. For a discussion of scaling analysis and citations to the original literature, see [4]. In the present context, it is a matter of balancing the inertial, stiffness, and (in-phase) aerodynamic forces. But there is computational evidence to support this as well.

And finally, as we see, if we divide this compliance ratio by the mass ratio, what is sometimes called the "reduced velocity" is obtained:

$$V \equiv U_\infty/\omega_R L \tag{3}$$

Conversely, the reduced velocity V is of order 1 for single-degree-of-freedom flutter. Thus, the compliance ratio λ^* for single-degree-of-freedom flutter is much less than the compliance ratio for coupled-mode flutter for typical small values of mass ratio μ. This is thought to be a major contributor to the precipitous decrease in the compliance ratio (in dimensional terms, the flow dynamic pressure), as there is a transition from coupled-mode flutter to single-degree-of-freedom flutter as the Mach number moves into the transonic range.

It turns out that coupled-mode flutter is strongly dependent on the compliance ratio but is weakly dependent on the mass ratio, while single-degree-of-freedom flutter is largely governed by the reduced velocity with a subtle dependence on the mass ratio, which determines the growth rate of single-degree-of-freedom flutter. Again, see [1,3].

V. Single-Degree-of-Freedom Flutter

It is known that, within the framework of potential flow theory (and also for the inviscid Euler equations even in the presence of shocks),

negative damping will only occur in a certain Mach number range and below a certain reduced frequency, where the reduced frequency is defined as follows:

$$k \equiv \frac{\omega L}{U_\infty} = \frac{\omega L}{a_\infty} \Big/ M_\infty \tag{4}$$

[As an aside, in turbomachinery flutter, single-degree-of-freedom flutter is much more common than coupled-mode flutter, while for airframes both can occur, although coupled-mode flutter is much more often studied. See chapters "Aeroelasticity in Turbomachines" and "Modern Analysis of Complex and Unsteady Flows in Turbomachines" in Ref. [5].]

Note that, for both wind tunnel and flight tests, this reduced frequency might be better expressed in terms of speed of sound rather than flow velocity. See Eq. (4). In the flight test, the speed of sound a_∞ is constant at constant altitude, while in the wind tunnel test, it is the stagnation speed of sound a_o that is constant. Thus, for a wind tunnel test, Eq. (4) may be rewritten as follows:

$$k = \left(\frac{\omega L}{a_o}\right)\left(\frac{a_o}{a_\infty}\right) \Big/ M_\infty \tag{5}$$

Equation (5) may further be rewritten as

$$\frac{\omega L}{a_0} = k\left(\frac{a_\infty}{a_0}\right) M_\infty \tag{6}$$

Note that at the single-mode flutter boundary, the reduced frequency, say, $k = k_F$, is a function of the Mach number. Thus, the right-hand side of Eq. (6) only depends on the Mach number. Thus, the criterion for preventing single-degree-of-freedom flutter is as follows:

$$\frac{\omega_s L}{a_o} > k_F\left(\frac{a_\infty}{a_o}\right) M_\infty \tag{7}$$

where ω_s is a structural resonant frequency.

A criterion like Eq. (7) is commonly used in turbomachinery and has also on occasion been used for control surface "buzz," which indeed, in its simplest form, is an example of single-degree-of-freedom flutter.

Since the flutter frequency is the same (or nearly the same) as a resonant frequency, to prevent single-degree-of-freedom flutter, the resonant frequency of each mode must be sufficiently large that the reduced frequency is in the (higher) stable range. Again, see Eq. (7). Note also that the critical reduced frequency will generally be different for each structural mode. Fortunately, only a few structural modes will be critical, and the Mach number range in which single-degree-of-freedom flutter may occur is limited. It is most likely to occur in the transonic/low supersonic range. Indeed, for simple wing shapes and skin panels, potential flow theory predicts that single-degree-of-freedom flutter will not occur above a Mach number equal to the square root of 2. Also, for low-aspect-ratio structures the Mach number range for single-degree-of-freedom flutter will be further reduced. Finally, single degree of freedom does not occur at low subsonic Mach numbers except for extreme geometries; e.g., the elastic axis is well ahead of the leading edge of an airfoil.

VI. Illustrative Example of Single-Mode Flutter

Consider a thin airfoil hinged at its leading edge with a spring and inertia to give it a resonant structural frequency. For supersonic flow, this can also be considered a control surface attached to an (immovable) airfoil. Single-degree-of-freedom flutter of a control surface is sometimes called "buzz" and may be complicated in practice because of a shock standing on the control surface or on the airfoil ahead of the hinge line. But single-degree-of-freedom flutter can occur even in the absence of a shock in low supersonic flow, as this example will show.

Taking the results from Fig. 2 [6] for the damping (out-of-phase component) of the aerodynamic moment, it is seen that for Mach numbers above 1.414, this moment component always produces positive damping, although the aerodynamic damping does go to zero as the reduced frequency goes to zero as it must. However, for $M < 1.414$ below a certain reduced frequency, the aerodynamic damping is negative. The reduced frequency at which the aerodynamic damping goes from positive to negative is the flutter condition expressed in terms of reduced frequency at a given Mach number.

The values of the flutter reduced frequencies at several Mach numbers are given below. Again, see Fig. 2: $k = 0.32$ at $M = 1.1$; $k = 0.53$ at $M = 1.2$; $k = 0.55$ at $M = 1.3$.

From these data, the left-hand side of Eq. (7) may be evaluated using the standard gas dynamic tables: $\omega_s L/a_o = 0.32$ at $M = 1.1$; $(\omega_s L/a_o) = 0.55$ at $M = 1.2$; $(\omega_s L/a_o) = 0.69$ at $M = 1.3$.

Thus, one might conclude that flutter is more likely when the Mach number increases because a higher natural frequency of the structure is required to prevent flutter. But recall that beyond $M = 1.414$ there is no single-degree-of-freedom flutter at all. Is this a paradox? Not really, for the reasons described next.

Although the natural or resonant structural frequency required to prevent any flutter is indeed going up (in the absence of any structural damping), the peak magnitude of the negative damping is going down as the Mach increases, and thus a smaller and smaller amount of structural damping is needed to offset the negative aerodynamic damping as the Mach number increases toward 1.414. See again Fig. 2.

Finally, the amount of structural damping required to prevent single-degree-of-freedom flutter scales in proportion to the fluid/structural mass ratio μ. This can be seen by writing the aeroelastic equation of motion and balancing the (positive) structural damping terms and the (negative) aerodynamic damping terms. However, an even simpler argument may suffice for present purposes. Structural damping models are normally written as a damping coefficient g times the structural stiffness, while the aerodynamic damping term is that portion of the aerodynamic (generalized) force (the moment in the present example) that is out of phase with the structural deflection (and thus in phase with the structural velocity). Thus, to a good approximation, where $C_{M_\dot{\alpha}}$ is the aerodynamic damping (the out-of-phase or imaginary part of the aerodynamic moment in the present example),

$$g m \omega_R^2 L^4 \sim \rho_\infty U_\infty^2 L^3 C_{M_\dot{\alpha}} k \tag{8}$$

or

$$g \sim \left(\frac{U_\infty}{\omega_R L}\right) C_{M_\dot{\alpha}} \mu \tag{9}$$

where the last result follows from Eq. (2). Recalling that the critical reduced velocity and reduced frequency are of order 1 for

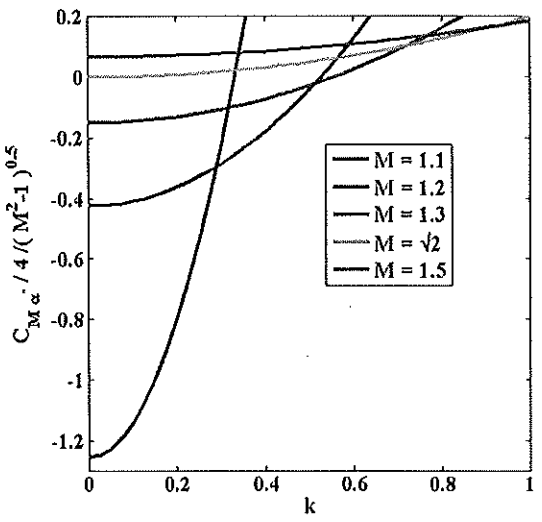


Fig. 2 Aerodynamic damping versus reduced frequency k for several Mach numbers.

single-degree-of-freedom flutter, this implies that the amount of structural damping to prevent such flutter scales with the mass ratio. This is another example of scaling analysis. Please note that Eqs. (8) and (9) are order of magnitude estimates and not strict equalities per se.

For typical values of reduced velocity, mass ratio, and aerodynamic coefficients, the required value for structural damping to prevent such flutter is much less than 1. Most importantly, the above discussion demonstrates, in a simple example, how sensitive the flutter boundary is to mass ratio and g. Also, it is true that most (all?) aerodynamic models determine the aerodynamic damping coefficients (out of phase with respect to structural motion) less accurately than they do the aerodynamic stiffness coefficients (in phase with respect to structural motion).

Single-degree-of-freedom flutter is more sensitive and subtle than coupled-mode flutter, and this simple example illustrates that. The sensitivity of single-degree-of-freedom flutter to the amount of structural damping does provide a possible way to decrease the likelihood of this type of flutter. This is unlike coupled-mode flutter, which is much less sensitive to the amount of structural damping.

VII. Coupled-Mode Flutter

The limit of potential flow theory as the Mach number becomes large is what has come to be called "piston theory." Indeed, a generalization of "piston theory" is available [6] as an expansion of potential flow theory as a series in the inverse of Mach number squared. For the present purposes, there are two features of this high-Mach-number limit that are highlighted here. One is that piston theory always provides positive aerodynamic damping, and therefore it demonstrates explicitly that single-degree-of-freedom flutter will not occur at high Mach numbers. Only coupled-mode flutter will occur at high Mach numbers.

But piston theory also shows that the three parameters discussed above may be reduced to two. The compliance ratio and mass ratio may each be divided (normalized) by the Mach number, and when this is done, the Mach number no longer appears as a separate parameter. Hence, in the limit of high Mach number, there are only two essential parameters, not three. Moreover, for typical small fluid/structural mass ratios, in this limit of high Mach number, the results will only be weakly dependent on mass ratio, and, indeed often (but not always), one may set the mass ratio to zero. This provides a conservative estimate of the flutter dynamic pressure for coupled-mode flutter. Thus, in the high-Mach-number limit, only coupled-mode flutter will occur, and only a single parameter governs the flutter phenomenon.

This was perhaps first observed in the flutter of plates and shells [7], but is true of wing flutter as well.

One caveat to be noted is that if the product of Mach number times either the angle of attack (in radians) or the thickness-to-chord ratio of the wing or curved panel becomes of order 1, then the airfoil profile and angle of attack may become important parameters at a high (hypersonic) Mach number.

VIII. Effect of Shocks and Viscous Boundary Layers

Shocks, of course, complicate the computations of flutter and other aeroelastic or fluid-structural interaction phenomena. But shocks per se in an inviscid flow do not change the broad conclusions drawn here. More specifically, if the Mach numbers ahead and behind the shock are high enough, one may use a local version of the piston theory with local flow properties, and, again, only coupled-mode flutter will occur.

The effect of a viscous boundary layer has been studied in the context of the flutter of plates, with some remarkable success in correlating theory and experiment. (See [7] and the chapter "Some Recent Advances in Nonlinear Aeroelasticity" in [5] for a discussion of the relevant literature.) It is clear that the effect of a viscous boundary layer is much larger for single-degree-of-freedom flutter than for coupled-mode flutter. As an approximate rule of thumb, if the ratio of the viscous boundary-layer thickness to plate length is less than 5%, there is little change in the flutter dynamic pressure for coupled-mode flutter. On the other hand, the dynamic pressure for single-degree-of-freedom flutter might change by 100% for such a boundary-layer thickness. This highlights again the greater challenge of correlating theory and experiment for single-degree-of-freedom flutter versus coupled-mode flutter. It is also known that single-mode flutter is much more sensitive to structural damping than coupled-mode flutter, as shown by the example included in this Note. As a very rough analogy, one may think of the viscous boundary layer as changing the dissipation in the aeroelastic or fluid structural system. However, the viscous boundary layer also changes the effective Mach number and dynamic pressure of the flow.

IX. Effects of Airfoil Profile and a Static Angle of Attack as Additional Parameters

Of course, shock strength, location, and viscous boundary layers are outcomes of parameters such as airfoil profile and/or a static angle of attack, as well as Mach number and Reynolds number. Raveh and the Technion team [8] and Stanford and the NASA Langley Research Center team [9] have done valuable computational studies of the Benchmark Wing and the AGARD 445.6 Wing exploring these additional parameters and have provided strong evidence of flutter due to negative aerodynamic damping in the transonic range, with flutter occurring at near-zero dynamic pressure. It is clear that these parameters can change the range of flow parameters where flutter due to negative aerodynamic damping may occur. Thus, in addition to the three fundamental flow parameters considered here, there is in fact a higher-dimensional parameter space, including airfoil profile and angle of attack, that needs to be considered, especially for single-degree-of-freedom flutter due to negative aerodynamic damping in transonic flow (and possibly in the hypersonic flow regime as well, as discussed previously).

X. Conclusions

In this brief Note, the importance of thinking of flutter as a three-parameter phenomenon rather than a two-parameter projection on this three-parameter space has been emphasized. The special importance of this for transonic flutter, where both coupled-mode flutter and single-degree-of-freedom flutter may occur, has been noted. And the implications for computational, wind tunnel, and flight investigation of flutter have been pointed out.

It is noted that in a moderately to highly complex aeroelastic system with several degrees of freedom, a pure coupled-mode flutter or a pure single-degree-of-freedom flutter is an idealization although a useful one. Moreover, there is the distinct possibility of a rapid change in the contributions of various structural modes to the most critical flutter boundary, especially when traversing the transonic flow range. And it is also the case that additional key nondimensional parameters may be important when static loadings due to the airfoil profile or static angle of attack induce nonlinear static effects that move the flutter boundary. Limit cycle oscillations are yet another step beyond flutter and will add other parameters to be considered.

An important consequence of this greater sensitivity of single-degree-of-freedom flutter (arising from negative aerodynamic damping) to a larger number of parameters than those for coupled-mode flutter is the need for highly efficient computational methods that can investigate all of the essential parameters over a wider range of values of interest. A systematic investigation of the sensitivity of the flutter boundary to these parameters will tell us much. The representation of the aerodynamic forces by a convolution integral in the time domain [10–12] or by an aerodynamic transfer function in the frequency domain [5,13] that can be used at any level of fluid flow modeling, from potential flow to Euler to RANS, are the most promising approaches for such computations.

Finally, the important topic of "buffet" [14] has not been addressed here. Buffet is here defined as the global dynamic instability of the flowfield, even in the absence of any structural motion. Of course, once the flow begins to oscillate in a limit cycle (called "buffet" in the airframe industry and "nonsynchronous vibration" in the

turbomachinery world), then the structure will respond and may indeed interact significantly with the flowfield. Buffet is an inherently nonlinear dynamic phenomenon and is not further discussed here, though it is of great interest to both the airframe and turbomachinery communities.

In the Appendix, an alternative nondimensional compliance parameter is presented that often appears in the literature, and it is related to that discussed in the main text.

For a stimulating review of transonic flutter that includes a discussion of the three-parameter issue, see the paper by Bendiksen [15]. Much of that paper deals with transonic similarity laws and the role of aerodynamic nonlinearities, but there is also an insightful discussion of the differences in conducting wind tunnel tests in freon versus air and the consequent differences in mass ratio.

Appendix: An Alternative Nondimensional Compliance Parameter

For skin panels and plate-like lifting surfaces typical of high-speed flight vehicles, another nondimensional compliance parameter often appears in the literature. This parameter can be related to the compliance parameter previously considered in this Note. The Mach number and mass ratio are the same. However, an alternative compliance ratio is often defined as follows, where D is the plate stiffness:

$$\lambda_D^* \equiv \rho_\infty U_\infty^2 L^3/D \tag{A1}$$

Similarly, a nondimensional frequency can be defined as follows:

$$\Omega_R^2 \equiv m\omega_R^2 L4/D \tag{A2}$$

Then the nondimensional compliance ratio defined in the main text can be shown to be the following:

$$\lambda^* = \lambda_D^*/\Omega_R^2 \tag{A3}$$

For typical structures, Ω_R is of the order of 10, and for coupled-mode flutter, λ_D^* is of the order of 100, so λ^* is of the order of 1 for coupled-mode flutter. See [7] for more details.

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